TWO INSTABILITY INTERVALS FOR A SWITCHED PLANAR SYSTEM WITH ONE NON-HURWITZ MATRIX

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ABSTRACT. Let $\dot{X}_t = A_{I_t} X_t$ in $\mathbb{R}^2$, where I jumps between 0 and 1 at rate r. Malrieu asked whether A_0, A_1 can be chosen so that the set of unstable rates is a union of several intervals. For the Hurwitz matrix $A_0 = \begin{pmatrix} -3/4 & -1 \\ 1 & -3/4 \end{pmatrix}$ and the non-Hurwitz matrix $A_1 = \operatorname{diag}(5/4, -11/4)$ we prove that the top Lyapunov exponent $\lambda(r)$ is continuous, positive on $(0, 1/10]$, negative on $[4.997, 5.003]$ and positive on $[19.95, 20.05]$, so the unstable set has at least two components. The middle signs come from trigonometric polynomials with exact rational coefficients that solve the Poisson equation of the angular process up to an explicit error. Numerically λ changes sign twice, near $r = 1.2228$ and $r = 7.7616$. The case in which both matrices are Hurwitz, as in the example behind the question, remains open.

1. INTRODUCTION

Let A_0, A_1 be real 2×2 matrices and $r > 0$. Let $(I_t)_{t \geq 0}$ be the Markov process on $\{0, 1\}$ that jumps from 0 to 1 and from 1 to 0 at rate r, and let X solve $\dot{X}_t = A_{I_t} X_t$. Equivalently, (X, I) is the Markov process on $\mathbb{R}^2 \times \{0, 1\}$ with generator

$$Lf(x, i) = (A_i x) \cdot \nabla_x f(x, i) + r\big(f(x, 1 - i) - f(x, i)\big). \tag{1}$$

This is the setting of Section 4.1 of Malrieu's survey [10], which presents the example of Benaïm, Le Borgne, Malrieu and Zitt [1]: there A_0 and A_1 are Hurwitz (all eigenvalues have negative real part), so is every convex combination of them, and yet $|X_t| \to \infty$ almost surely for some rates r. For that pair the set of unstable rates is empty or, conjecturally, an interval [10, Conjecture 4.3]. Malrieu then asks the following.

Open question 3 of [10] (Oscillations of the Lyapunov exponent). Is it possible to choose the two 2×2 matrices A_0 and A_1 in such a way that the set of jump rates r associated to unstable processes is the union of several intervals?

The question is stated for two real 2×2 matrices. Theorem 1.1 answers it with a pair in which A_0 is Hurwitz and A_1 is not; for two Hurwitz matrices it is open (Section 6). Let $\lambda(r)$ be the top Lyapunov exponent of the switched system, defined in Section 2.

Theorem 1.1. *Let*

$$A_0 = \begin{pmatrix} -3/4 & -1 \\ 1 & -3/4 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 5/4 & 0 \\ 0 & -11/4 \end{pmatrix}. \tag{2}$$

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Then λ is continuous on $(0,\infty)$, and for every $X_0 \neq 0$ and every $r > 0$, $t^{-1}\log|X_t| \to \lambda(r)$ almost surely. Moreover

$$\begin{aligned} \lambda(r) &> \tfrac{9}{50} && \text{for } 0 < r \le \tfrac{1}{10}, \\ -0.0756 \le \lambda(r) &\le -0.0055 && \text{for } 4.997 \le r \le 5.003, \\ 0.0361 \le \lambda(r) &\le 0.1192 && \text{for } 19.95 \le r \le 20.05. \end{aligned}$$

Consequently the set $U = \{r > 0 : \lambda(r) > 0\}$ of unstable rates is open and has at least two connected components, one containing $(0, 1/10]$ and another containing $[19.95, 20.05]$, and they are separated by the stable interval $[4.997, 5.003]$.

The matrix A_0 has eigenvalues $-3/4 \pm i$ and is Hurwitz; A_1 has eigenvalues $5/4$ and $-11/4$ and is not. The average $\bar{A} = (A_0 + A_1)/2$ has the positive eigenvalue $-3/4 + \sqrt{3}/2$. Malrieu defines the unstable rates as those for which $|X_t| \to \infty$ almost surely. By Theorem 1.1 this set contains $(0, 1/10]$ and $[19.95, 20.05]$ and misses $[4.997, 5.003]$, so it is not an interval under either definition.

The bounds at $r = 5$ and $r = 20$ come from explicit test functions $f = (f_0, f_1)$ on the projective line. Write $X_t = |X_t| u_{\Theta_t}$; the pair (Θ, I) is the *angular process*, and $\mathcal{L}_r$ is its generator. If $h_i + (\mathcal{L}_r f)_i$ lies in $[\ell, u]$ everywhere, where h_i is the radial growth rate in state i, then $\ell \le \lambda(r) \le u$ (Lemma 3.1). The equation $h_i + (\mathcal{L}_r f)_i = \lambda(r)$ is the Poisson equation of the angular process, and a pair f with rational coefficients for which these bounds are computed exactly is called a *certificate* at rate r. We take f_i to be trigonometric polynomials of degree $2K$ with rational coefficients, $K = 128$ at $r = 5$ and $K = 32$ at $r = 20$; then $h_i + (\mathcal{L}_r f)_i$ is again a trigonometric polynomial, its Fourier coefficients are rational numbers computed exactly from those of f, and its range is bounded by the sum of their absolute values.

Section 2 sets up the angular process and proves that λ exists and is continuous. Section 3 proves the certificate lemma and the signs near $r = 5$ and $r = 20$, and Section 4 the sign at small rates together with the proof of Theorem 1.1. Section 5 computes the whole curve from the stationary density, and Section 6 discusses the case of two Hurwitz matrices.

Related work. Lawley, Mattingly and Reed [9] prove that switching between Hurwitz matrices is stable at slow and fast rates when the averaged matrix is Hurwitz. In their Example 3.3 they take direct sums of planar blocks to build pairs of $2k \times 2k$ matrices whose unstable set has k separated components. The smallest case with two components is in dimension four. Lagasquie [8] looks for a second component of the unstable set in the planar setting of [1] and finds one after replacing the exponential holding times by Gamma holding times; his model keeps two planar matrices but changes the switching law. Porfiri, Jeter and Belykh find disconnected windows of stability for jump linear systems in discrete time, where an independent matrix is held for an integer number of steps [14], and antiresonance in a nonlinear planar system that switches between unstable modes [15]. Neither setting is the two-state exponential switching of (1). Benaïm and Strickler [3] relate the top exponent of switched linear systems to invariant measures of the angular process and treat fast switching; Chitour, Mazanti, Monmarché and Sigalotti [4] compare deterministic and probabilistic exponents and give the fast-switching limit used in Section 5. Slow and fast limits for cooperative systems are in [2] and [12], and a three-dimensional cooperative example with a stable convex hull and unstable switching is in [13]. None of these gives two planar matrices whose unstable set has two components under the switching law (1).

2. The angular process and the exponent

For $\theta \in \mathbb{R}/\pi\mathbb{Z}$ put $u_\theta = (\cos\theta, \sin\theta)$ and $u_\theta^\perp = (-\sin\theta, \cos\theta)$. For a matrix A let $h_A(\theta) = u_\theta^{\mathsf{T}} A u_\theta$ and $v_A(\theta) = (u_\theta^\perp)^{\mathsf{T}} A u_\theta$. Writing $X_t = R_t u_{\Theta_t}$ with $R_t = |X_t|$, the equation $\dot X = A_I X$ becomes

$$\frac{d}{dt}\log R_t = h_{I_t}(\Theta_t), \qquad \dot\Theta_t = v_{I_t}(\Theta_t), \tag{3}$$

with $h_i = h_{A_i}$ and $v_i = v_{A_i}$. The angular process (Θ, I) is a Markov process on $(\mathbb{R}/\pi\mathbb{Z}) \times \{0,1\}$ whose generator acts on pairs $f = (f_0, f_1)$ of C^1 functions with period π by

$$(\mathcal{L}_r f)_i = v_i f_i' + r(f_{1-i} - f_i). \tag{4}$$

For the matrices (2),

$$v_0 = 1, \qquad v_1(\theta) = -2\sin 2\theta, \qquad h_0 = -\tfrac{3}{4}, \qquad h_1(\theta) = -\tfrac{3}{4} + 2\cos 2\theta. \tag{5}$$

In state 0 the direction rotates at unit speed and the norm decays at rate $3/4$; in state 1 the direction moves toward the horizontal axis.

Proposition 2.1. *For the matrices* (2) *and every* $r > 0$ *there is* $\lambda(r) \in \mathbb{R}$ *such that* $t^{-1}\log|X_t| \to \lambda(r)$ *almost surely for every* $X_0 \neq 0$ *and every initial state* I_0*;* $\lambda(r)$ *is the top Lyapunov exponent of the matrix cocycle, and* λ *is continuous on* $(0,\infty)$*.*

Proof. Start the clock at a jump into state 0 and call a *cycle* a holding period T_0 in state 0 followed by a holding period T_1 in state 1; the holding times are independent and exponential with mean $1/r$. Let $\Theta^{(n)}$ be the direction at the start of the n-th period in state 1. During a period in state 0 the direction moves by T_0 modulo π, and T_0 modulo π has density

$$k_r(s) = \frac{re^{-rs}}{1 - e^{-\pi r}}, \qquad 0 \le s < \pi, \tag{6}$$

so $m_r \le k_r \le M_r$ with $m_r = r/(e^{\pi r} - 1)$ and $M_r = r/(1 - e^{-\pi r})$. Since the period in state 1 is independent of the next period in state 0, the transition kernel of the chain $(\Theta^{(n)})$ is bounded below by $m_r\, d\theta$, uniformly in the starting direction. By Doeblin's condition the chain has a unique stationary law, converges to it in total variation at a geometric rate uniform in the starting point, and its ergodic averages converge almost surely from every starting point [11, Theorem 16.0.2].

The increment of $\log R$ over a cycle is a function of the direction at its start and of (T_0, T_1), bounded in absolute value by $\|A_0\|T_0 + \|A_1\|T_1$, which has finite second moment. The ergodic theorem for the Markov chain formed by the directions and the holding times, together with the strong law for $\sum(T_0 + T_1)$, gives $t^{-1}\log R_t \to \lambda(r)$ along cycle ends, with $\lambda(r)$ equal to the stationary mean of the cycle increment divided by the mean cycle length $2/r$. With $C = \max_i \|A_i\|$, between cycle ends $\log R$ moves by at most $C(T_0 + T_1)$ for the current cycle, and $\max_{n \le N}(T_0^{(n)} + T_1^{(n)})/N \to 0$ almost surely, so the limit holds along all t. It does not depend on $X_0 \neq 0$; applying it to two independent initial vectors identifies $\lambda(r)$ with the top exponent of the cocycle.

For continuity, fix a compact interval $J \subset (0,\infty)$. The constant m_r is bounded below on J, so the Doeblin bound is uniform there. The holding times have joint density $r^2 e^{-r(t_0+t_1)}$, continuous in r in L^1, while the deterministic maps from the holding times to the next direction and to the cycle increment do not depend on r. Hence the transition kernel is continuous in r in total variation, uniformly in the starting direction, and by the uniform geometric contraction so is the stationary law. The conditional mean of the cycle increment given the starting direction

is bounded and continuous in r uniformly in the direction, by dominated convergence with the bound $C(t_0+t_1)$. Dividing the stationary mean by $2/r$ gives continuity of λ on J. □

3. Certificates

Lemma 3.1. *Let $r>0$ and let $f=(f_0,f_1)$ be a pair of C^1 functions of period π. Put $g_i=h_i+(\mathcal{L}_r f)_i$. If $\ell\le g_i(\theta)\le u$ for $i=0,1$ and all θ, then $\ell\le\lambda(r)\le u$.*

Proof. By Dynkin's formula, $M_t=f_{I_t}(\Theta_t)-f_{I_0}(\Theta_0)-\int_0^t(\mathcal{L}_r f)_{I_s}(\Theta_s)\,ds$ is a martingale. Its jumps occur at the jumps of I, at rate r, and are bounded by $2\|f\|_\infty$, so $\mathbb{E}M_t^2\le 4r\|f\|_\infty^2 t$ and $M_t/t\to 0$ almost surely by the strong law for square-integrable martingales [5, Theorem 2.18] applied along integer times, together with the bound on the jumps in between. By (3),

$$\frac{1}{t}\log\frac{R_t}{R_0}=\frac{1}{t}\int_0^t g_{I_s}(\Theta_s)\,ds-\frac{f_{I_t}(\Theta_t)-f_{I_0}(\Theta_0)-M_t}{t},$$

and the last term tends to 0. The integral average lies in $[\ell,u]$, and the left side tends to $\lambda(r)$ by Proposition 2.1. □

Take $f_i(\theta)=c_{i,0}+\sum_{k=1}^{K}\bigl(c_{i,k}\cos 2k\theta+s_{i,k}\sin 2k\theta\bigr)$ with rational coefficients. Since $v_0=1$ and $v_1=-2\sin 2\theta$, each g_i is a trigonometric polynomial of degree at most $2(K+1)$ with rational coefficients,

$$g_i(\theta)=C_i+\sum_{k=1}^{K+1}\bigl(\alpha_{i,k}\cos 2k\theta+\beta_{i,k}\sin 2k\theta\bigr), \tag{7}$$

and therefore

$$C_i-\sum_{k=1}^{K+1}\bigl(|\alpha_{i,k}|+|\beta_{i,k}|\bigr)\le g_i\le C_i+\sum_{k=1}^{K+1}\bigl(|\alpha_{i,k}|+|\beta_{i,k}|\bigr). \tag{8}$$

Proposition 3.2. *There are rational pairs $f^{(5)}$ with $K=128$ and $f^{(20)}$ with $K=32$, all coefficients with denominators dividing 10^{12}, for which the bounds (8) at $r=5$ and $r=20$ are the numbers in Table 1. Consequently*

$$-0.071355633951\le\lambda(5)\le-0.009745897257,\qquad 0.070309824390\le\lambda(20)\le 0.084999667350.$$

r	state i	lower bound in (8)	upper bound in (8)
5	0	−0.040550774017	−0.040550757183
5	1	−0.071355633951	−0.009745897257
20	0	0.077654745278	0.077654746482
20	1	0.070309824390	0.084999667350

Table 1. Exact ranges of $g_i=h_i+(\mathcal{L}_r f)_i$ for the two certificates. Every entry is an exact rational with denominator dividing 10^{12}, written in full.

Proof of Proposition 3.2. The coefficients are listed in the repository, in the file `data/poisson_certificates.json`. They are twelve-place decimal roundings of a floating-point solution of the Poisson equation restricted to the harmonics $k\le K$. Given the coefficients, (7) is computed by exact rational arithmetic, using $\frac{d}{d\theta}\cos 2k\theta=-2k\sin 2k\theta$, $\frac{d}{d\theta}\sin 2k\theta=2k\cos 2k\theta$ and the product formulas for $\sin 2\theta\cos 2k\theta$ and $\sin 2\theta\sin 2k\theta$. The file `code/verify_certificate.py` carries out this computation with exact rationals in the basis $\cos 2k\theta,\sin 2k\theta$, and `code/`

`independent_certificates.py` in the basis z^k, $z = e^{2i\theta}$, where $v_1 = i(z - z^{-1})$; both give the entries of Table 1. Lemma 3.1 with $\ell = \min_i$ and $u = \max_i$ of the table entries gives the bounds. □

Corollary 3.3. $-0.0756 \le \lambda(r) \le -0.0055$ *for* $4.997 \le r \le 5.003$ *and* $0.0361 \le \lambda(r) \le 0.1192$ *for* $19.95 \le r \le 20.05$.

Proof. By (4), $(\mathcal{L}_r f)_i - (\mathcal{L}_{r_0} f)_i = (r - r_0)(f_{1-i} - f_i)$, so with the same f the range of g_i at rate r is that at r_0 widened by $|r - r_0|\,\|f_1 - f_0\|_\infty$ on each side. Bounding $\|f_1 - f_0\|_\infty$ by the sum of the absolute values of the Fourier coefficients of $f_1 - f_0$ gives at most 1.3918 for $f^{(5)}$ and at most 0.6824 for $f^{(20)}$ (the exact rationals are in `data/certified_neighborhoods.json`). For $f^{(5)}$ the widened ranges of g_0 and g_1 lie in

$$[-0.071355633951 - 0.003 \cdot 1.3918,\ -0.009745897257 + 0.003 \cdot 1.3918] \subset [-0.0756,\ -0.0055],$$

and for $f^{(20)}$ they lie in

$$[0.070309824390 - 0.05 \cdot 0.6824,\ 0.084999667350 + 0.05 \cdot 0.6824] \subset [0.0361,\ 0.1192],$$

since by Table 1 the range of g_0 lies inside that of g_1 at both rates. Lemma 3.1 gives the bounds. □

4. Small rates

Proposition 4.1. *For every* $r > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{\pi \log 2}{2} \, \frac{r^2}{1 - e^{-\pi r}}. \tag{9}$$

In particular $\lambda(r) > 9/50$ *for* $0 < r \le 1/10$, *and* $\lambda(r) \to 1/4$ *as* $r \to 0$.

Proof. Use the cycles of the proof of Proposition 2.1. In state 0, $|e^{A_0 t} x| = e^{-3t/4}|x|$. In state 1, for a unit vector u_θ,

$$\log |e^{A_1 T_1} u_\theta| \ge \log\bigl(e^{5T_1/4} |\cos\theta|\bigr) = \tfrac{5}{4} T_1 + \log|\cos\theta|.$$

The direction $\Theta^{(n)}$ at the start of a period in state 1 is the previous direction shifted by an independent T_0 modulo π, so its conditional density given the past is at most M_r. Since $\log|\cos\theta| \le 0$ and $\int_0^\pi \log|\cos\theta|\, d\theta = -\pi \log 2$,

$$\mathbb{E}\bigl[\log|\cos\Theta^{(n)}| \,\big|\, \text{past}\bigr] \ge -M_r \pi \log 2.$$

The stationary mean of the cycle increment is therefore at least $-\frac{3}{4r} + \frac{5}{4r} - M_r \pi \log 2$, and dividing by the mean cycle length $2/r$ gives (9). Using $1 - e^{-x} \ge x/(1+x)$ for $x > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{r(1 + \pi r)\log 2}{2},$$

and for $0 < r \le 1/10$, with $\pi < 4$ and $\log 2 < 1$, the right side exceeds $\frac{1}{4} - \frac{1}{20} \cdot \frac{14}{10} = \frac{9}{50}$. For the limit, the cycle increment is at most $-\frac{3}{4} T_0 + \frac{5}{4} T_1$, so $\lambda(r) \le \frac{1}{4}$ for every r. □

Proof of Theorem 1.1. Continuity and the almost sure limit are Proposition 2.1. The bounds are Proposition 4.1 and Corollary 3.3. Since λ is continuous, U is open in $(0, \infty)$ and each of its connected components is an open interval. The points $1/10$ and 20 lie in U, and the point 5 between them does not, so they lie in different components. □

5. Numerical picture

The certificates prove signs at chosen rates; the shape of the whole curve is computed numerically. The stationary law of (Θ, I) has densities p_0, p_1 satisfying $p_0' = r(p_1 - p_0)$ and $(v_1 p_1)' = r(p_0 - p_1)$, with constant positive flux $J = p_0 + v_1 p_1$ and $\int_0^\pi p_0 = \int_0^\pi p_1 = \frac{1}{2}$; then

$$\lambda(r) = -\frac{3}{4} + 2\int_0^\pi \cos(2\theta)\, p_1(\theta)\, d\theta.$$

With $H(\theta) = \theta - \frac{1}{4}\log|\tan\theta|$ (so that $H' = 1/v_0 + 1/v_1$) the solution is explicit on each quarter-circle: for $0 < \theta < \pi/2$,

$$F_-(\theta) = e^{-rH(\theta)} \int_\theta^{\pi/2} e^{rH(s)}\, ds, \qquad p_0 = J(1 + rF_-), \qquad p_1 = \frac{JrF_-}{2\sin 2\theta},$$

and for $\pi/2 < \theta < \pi$,

$$F_+(\theta) = e^{-rH(\theta)} \int_{\pi/2}^\theta e^{rH(s)}\, ds, \qquad p_0 = J(1 - rF_+), \qquad p_1 = -\frac{JrF_+}{2\sin 2\theta},$$

with J fixed by normalization. On the second branch $H' > 1$, which gives $rF_+ < 1$ and $p_0 > 0$. We evaluate these integrals as ordinary differential equations in the coordinate $x = \log|\tan\theta|$ on $(0, \pi/2)$ and $x = -\log|\tan\theta|$ on $(\pi/2, \pi)$, so that $\cos 2\theta = -\tanh x$ on the first branch and $\tanh x$ on the second, which resolves the power singularities at the attracting and repelling directions of A_1 (`code/candidate_density.py`). The same integrals computed in a second coordinate (`code/independent_density.py`) give values of λ that differ from these by less than 10^{-12} at $r = 1/10$, 5 and 20. Table 2 compares the density method with two other estimates at those rates.

r	stationary density	Markov chain, 16384 cells	Monte Carlo (standard error)
0.1	0.216444855966	0.216417929898	0.21600 (0.00036)
5	−0.040611505820	−0.040711536760	−0.04159 (0.00066)
20	0.077660185766	0.077491305516	0.07702 (0.00061)

Table 2. Three estimates of $\lambda(r)$. The Markov chain discretizes the angle into cells with upwind moves at rate $|v_i|/\Delta\theta$; its remaining bias is mesh error. The Monte Carlo runs use exact matrix flows between exponential holding times, two seeds of 2×10^6 cycles each; the standard errors are batch estimates.

On a grid of 403 rates in $[10^{-3}, 10^2]$ the density method gives exactly two sign changes, near $r = 1.2228$ and $r = 7.7616$ (to ten places 1.2228361663 and 7.7615867624; Figure 1). A grid of 151 rates in $[10^{-3}, 10^3]$ gives the same two sign changes. The two crossings and the absence of others are numerical observations. As $r \to 0$, $\lambda(r) \to \frac{1}{2}(-\frac{3}{4} + \frac{5}{4}) = \frac{1}{4}$ by Proposition 4.1. As $r \to \infty$, $\lambda(r)$ tends to the largest eigenvalue $-\frac{3}{4} + \frac{\sqrt{3}}{2} \approx 0.1160$ of $\bar{A}$; this follows from [4, Proposition 6.4], whose condition (C) holds because the lower eigenspace of $\bar{A}$ is a line and A_0 leaves no real line invariant (their redraw intensity μ is $2r$ in our notation). So both ends of the curve are unstable and the exponent dips below zero in between.

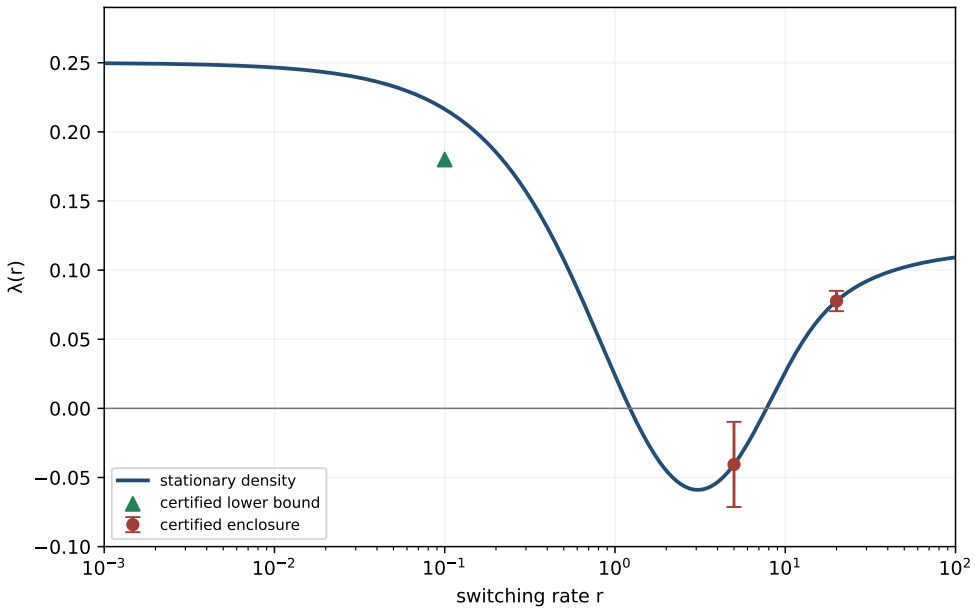


FIGURE 1. The exponent $\lambda(r)$ computed from the stationary density, on a logarithmic rate axis. The marked points are the rates where the sign is proved.

6. The Hurwitz case

In the example of [1] that precedes Open question 3 both matrices are Hurwitz, and one may read the question as asking for such a pair. Whether two Hurwitz matrices can have an unstable set with two components is open. Two remarks bear on it.

First, no pair of Hurwitz matrices of the form $A_0 - c_0\mathrm{Id}$, $A_1 - c_1\mathrm{Id}$ is unstable at any rate. Replacing A_0, A_1 by $A_0 - c_0\mathrm{Id}, A_1 - c_1\mathrm{Id}$ lowers λ by $(c_0 + c_1)/2$ for every r, because the two states are occupied half of the time each. Making A_1 Hurwitz needs $c_1 > 5/4$, and then $\lambda(r) < \frac{1}{4} - \frac{5}{8} < 0$ for every r by the bound $\lambda \le \frac{1}{4}$ in the proof of Proposition 4.1. More generally, a normal matrix has symmetric part with eigenvalues equal to the real parts of its own eigenvalues, so a normal Hurwitz matrix A satisfies $x^{\mathsf{T}}Ax \le -\epsilon|x|^2$ for some $\epsilon > 0$. If both matrices are normal and Hurwitz, then $\frac{d}{dt}|X_t|^2 \le -2\epsilon|X_t|^2$ along every switching path, and $\lambda(r) \le -\epsilon$ for every r. A Hurwitz pair whose unstable set has two components must therefore contain a non-normal matrix, as the pair of [1] does.

Second, among 1545 Hurwitz pairs, obtained by shifting each matrix of 515 random planar pairs to spectral abscissa $-\gamma$ for $\gamma \in \{0.01, 0.05, 0.1\}$, none has two components of the unstable set at the sampled rates (`data/hurwitz_search.json`). The exponents were computed with a 256-cell discretization of the angle at 49 rates in $[10^{-3}, 10^{3}]$. Components narrower than the spacing of the sampled rates are not resolved.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [19] and Math Brain [17]. A Codex agent (OpenAI) found the pair (2), constructed the certificates and proved Theorem 1.1; a second Codex agent checked the proofs and recomputed the certificates, the stationary density and the Monte Carlo estimates with its own programs; a Claude agent (Anthropic) ran a third discretization and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

The certificates, the programs that check them and the numerical computations are in the repository

`https://github.com/yspennstate/planar-switching-two-instability-intervals`.

The exact checks need only the Python standard library; the interval audit uses `mpmath` [7], and the numerical computations use NumPy [6] and SciPy [16].

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Appendix A. Notation

Letters used only inside one proof are not listed.

Symbol	Meaning	First appears
X_t, I_t	the planar component and the switching state	Section 1
A_0, A_1	the two matrices; from Theorem 1.1 on, the pair (2)	Section 1
r	the rate of each jump of I	Section 1
$\lambda(r)$	the top Lyapunov exponent	Theorem 1.1
U	the set of rates with $\lambda(r) > 0$	Theorem 1.1
$\bar{A}$	$(A_0 + A_1)/2$	Section 1
u_θ, $u_\theta^\perp$	the unit vector at angle θ and its rotation by $\pi/2$	Section 2
R_t, Θ_t	$\lvert X_t\rvert$ and the direction of X_t modulo π	(3)
h_i, v_i	radial growth rate and angular velocity in state i	(3)
T_0, T_1	holding times in states 0 and 1 during one cycle	Proposition 2.1
$\Theta^{(n)}$	the direction at the start of the n-th period in state 1	Proposition 2.1
k_r	the density of T_0 modulo π	(6)
$\mathcal{L}_r$	generator of the angular process (Θ, I)	(4)
m_r, M_r	lower and upper bounds of the density (6)	Proposition 2.1
$f = (f_0, f_1)$, g_i	a test pair and $h_i + (\mathcal{L}_r f)_i$	Lemma 3.1
K	the number of harmonics in a certificate	(7)
C_i, $\alpha_{i,k}$, $\beta_{i,k}$	Fourier coefficients of g_i	(7)
$f^{(5)}$, $f^{(20)}$	the certificates at $r = 5$ and $r = 20$	Proposition 3.2
p_0, p_1, J	stationary densities of (Θ, I) and their flux	Section 5
H, $F_\pm$	the functions in the explicit stationary density	Section 5

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